Introduction to Software Engineering

Project Proposal & Plan

UniEvent is a university event management platform integrating event approval, ticketing, payment confirmation, QR-based attendance, and event monitoring. This document presents the project's vision, scope, proposed solution, development plan, risks, and collaboration approach.

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Project Proposal & Plan

UniEvent – Team 07

Revision History

Version	Date	Changed by	Section(s)	Change
0.1	19/10/2026	Thái Quang Huy	2.1–2.4	First draft of the Vision: problem statement, position statement, stakeholders and MoSCoW feature list F01–F16.
0.1	19/10/2026	Trần Vũ Anh Kiệt	3.1–3.3	First draft of the C4 System Context diagram, technology stack with reasons and operating environment.
0.1	19/10/2026	Lê Nguyễn Quốc Học	4.1–4.6	First draft of phase gates, WBS Levels 1–3, estimates, Gantt chart, roles and cost estimate.
0.1	19/10/2026	Nguyễn Văn Minh	5, 6	First draft of the risk register R1–R9 and the tools and collaboration setup.
0.2	20/10/2026	Lê Nguyễn Quốc Học	2	Reviewed the Vision; review comments resolved by the section owner.
0.2	20/10/2026	Nguyễn Văn Minh	3	Reviewed the Proposed Solution; review comments resolved by the section owner.
0.2	20/10/2026	Trần Vũ Anh Kiệt	4	Reviewed the Project Plan; review comments resolved by the section owner.

Version	Date	Changed by	Section(s)	Change
0.2	20/10/2026	Thái Quang Huy	5, 6	Reviewed Risk Management and Tools; review comments resolved by the section owner.
0.3	20/10/2026	Thái Quang Huy	2.1, 2.3	Added actor-specific problems, the actor × feature matrix and the key-data stewardship table.
0.3	20/10/2026	Thái Quang Huy	7	Consolidated member AI usage entries into the AI Usage Log.
0.4	21/10/2026	Lê Nguyễn Quốc Học	2.4, 3.3	Limited PA5 payments to one authorised payOS channel with a mock adapter for CI.
0.4	21/10/2026	Lê Nguyễn Quốc Học	4.3–4.6	Reconciled capacity, contributor hours, target schedule and costs.
0.4	21/10/2026	Nguyễn Văn Minh	5, 6	Aligned R2–R4 and CI deadlines with Section 4; recorded the 21/10 setup verification.
1.0	21/10/2026	Thái Quang Huy	2.1–2.4	Split the problem and position statements per problem and per user; restored organisation administration in F01; added an actor column to the feature list.
1.0	21/10/2026	Lê Nguyễn Quốc Học	4.2–4.6	One executing owner per work package; added WBS 2.2.8 and 5.2.13; calendar dates; corrected the Table 6 total.
1.0	21/10/2026	Lê Nguyễn Quốc Học	All	Integrated, froze and submitted PA1; committed the PDF and source to /pa/PA1.

Objectives

This document focuses on the following topics:

- ✔ Completing the Project Proposal & Plan document with the following sections:
  - Vision

- Proposed Solution
- Project Plan
- Risk Management
- Tools & Collaboration Setup

✔ Including an AI usage log for this phase

✔ Understanding the Project Proposal & Plan document.

1 Member Contribution Assessment

Contributions must add up to 100%. For PA1 document sections, evidence may include the shared Google Docs version history, review comments and the PA1 GitHub Project issue. Individual section pull requests are not required by the team's co-editing workflow.

ID	Name	Role(s)	Contribution (%)	Evidence
24127044	Lê Nguyễn Quốc Học	Team lead / PM; Developer	25%	PA1 issue #31; W3 assignment/review record
24127177	Thái Quang Huy	Business analyst; Developer	25%	PA1 issue #29; W3 assignment/review record
24127196	Trần Vũ Anh Kiệt	Designer / Architect; Developer	25%	PA1 issue #30; W3 assignment/review record
24127205	Nguyễn Văn Minh	Tester / DevOps; Developer	25%	PA1 issue #32; W3 assignment/review record
	Total		100%	

Contributions total 100%. All four members confirmed the 25% allocation on 21/10/2026 in the Discord #decisions channel. Evidence per member: the section owned and the section reviewed in W3-Status.md, PA1 issues #29–#32, the Google Docs version history and the Revision History above.

Assignment and review record: W3-status.md

2 Vision

2.1 Preliminary Problem Statement

Business context

Student clubs, the Youth Union (Đoàn Thanh niên), and the Student Association (Hội Sinh viên) organise various campus events, including workshops, seminars, competitions, performances, and volunteer activities. Based on the team's initial experience, some of these events use free or low-cost tickets, while attendance records may contribute to students' conduct-score assessments (điểm rèn luyện), depending on university regulations. These practices will be further validated with student organisers during the requirements phase (PA2).

Current way of working

In our experience, an event is often run with a chain of unconnected tools. The organising committee sends the event plan to the Youth Union by e-mail or on paper. After approval, students register through a Google Form and pay by bank transfer with a free-text transfer description. Organisers reconcile bank statements against form responses by hand and then e-mail confirmations one by one. At the door, volunteers look names up in a spreadsheet or on a printed list. After the event, the attendance list is retyped and sent to the Youth Union as evidence for conduct score.

Problems. The main challenge is not the absence of individual tools for registration, payment, approval or attendance, but the fragmentation of these activities across disconnected systems: information is copied and reconciled by hand, which costs effort, causes errors and gives nobody a view of the whole event life cycle. The six problems below state who is affected and what a successful solution must achieve.

P1	Manual payment reconciliation

The problem of	matching bank transfers with free-text descriptions against Google Form responses by hand
affects	club organisers who handle registration and finance, and students who pay for tickets
the impact of which is	slow confirmation, missing, duplicated or mismatched payments, confirmations e-mailed one by one, and disputes about whether a student has paid
a successful solution would	create a VietQR payment link per reservation, confirm payment from a verified payOS webhook, issue exactly one ticket per paid order and flag exceptions for review (F05–F06)

P2	Unreliable ticket validation
The problem of	admitting attendees by searching spreadsheets or printed lists, without a centrally validated one-time ticket
affects	check-in volunteers, organisers and attendees waiting at the entrance
the impact of which is	queues at the door, admission with forwarded screenshots or duplicate entries, and inconsistent attendance records
a successful	issue signed QR tickets and validate them on the server from a phone browser, rejecting invalid and used tickets and recording check-in and check-out separately

solution would	(F06–F07)

P3	Opaque event approval
The problem of	sending event plans to the Youth Union or Student Association by e-mail or on paper
affects	organisers who submit events and the officers who approve them
the impact of which is	organisers cannot see the status of a request, officers keep no searchable history of decisions and reasons, and events are delayed by follow-up messages
a successful solution would	provide an in-system submission and approval queue with approve, reject-with-reason and request-changes decisions and a complete decision history (F02–F03)

P4	Weak attendance evidence
The problem of	collecting attendance on sign-in sheets and retyping it after the event
affects	attendees whose conduct score depends on it, organisers who report it and officers who verify it
the impact of	incomplete, hard-to-verify records, conduct-score appeals that are hard to settle,

which is	and repeated retyping work
a successful solution would	record attendance from QR check-in and check-out, export it as CSV (F07–F08) and optionally prepare draft conduct-score evidence for bulk review (F09)

P5	No real-time view of the event
The problem of	having no single live view of ticket sales, payments and check-ins
affects	organisers and their event leads
the impact of which is	capacity and staffing decisions based on outdated spreadsheets, with overselling or empty seats discovered too late
a successful solution would	show tickets sold, revenue and check-in rate in real time with CSV export (F08), backed by atomic capacity control (F05)

P6	Unmanaged organisations and privileges
The problem of	sharing form ownership, spreadsheets and payment details informally, with no register of organisations and their organisers
affects	the system administrator, organisers, approvers and every attendee whose personal

	data is collected
the impact of which is	the wrong people can edit or export participant data, nobody can show who granted access, and approvers cannot tell which club a submitter represents
a successful solution would	let administrators register organisations, grant organiser and approver roles scoped to an organisation, lock accounts and inspect an audit log, while organisers assign check-in staff to their own events (F01)

Evidence. These observations come from the first-hand experience of our team lead, Lê Nguyễn Quốc Học, a member of the Student Volunteer Team (Đội Sinh viên tình nguyện) who has helped organise several of its activities and attended many other university events. He confirms that the steps above describe how such activities are run today. We treat them as preliminary domain knowledge, not as statistical evidence about all student organisations. To reduce reliance on a single organisation, we will validate key workflow assumptions through discussions with additional student organisers and review publicly available event information where relevant during PA2 (WBS 2.1.2). We also plan to ask the Student Volunteer Team to pilot UniEvent.

Actor-specific problems and value (initial hypotheses; PA2 interviews must validate differences across organisations)

Actor / user group	Current task / data	Initial pain point	Problems	Planned UniEvent value	Validation status
Student attendee	Find an event; register; transfer the fee; show proof of attendance	Registration, payment and ticket confirmation span unconnected tools; attendance evidence may be lost.	P1, P2, P4	One reservation, one verified ticket, visible attendance result (F05–F07).	Team experience only; interview students in PA2.

Actor / user group	Current task / data	Initial pain point	Problems	Planned UniEvent value	Validation status
Club organiser	Submit an event; manage capacity, orders and the on-site team	Manual approval tracking, reconciliation and attendance reporting; no live status.	P1, P3, P4, P5	Event workflow, ticket configuration, payment status, check-in staff assignment, dashboard and CSV (F01–F06, F08).	Confirm approval, refund and reconciliation variants with organisers.
Check-in volunteer	Validate tickets at the venue gate	Spreadsheet lookup is slow; duplicate admission is hard to detect.	P2	Phone scanner limited to assigned events; server-side atomic validation; separate check-in and check-out (F07).	Test Android/iPhone, low connectivity and concurrent scanners.
Youth Union / Student Association approver	Review proposals from their unit; audit prior decisions	Decisions and reasons are fragmented and hard to audit.	P3, P4	Approval queue for their unit, reasoned decisions and history (F03); optional attendance review (F09).	Verify delegated authority and approval stages with officers.
System administrator	Register organisations; grant privileged roles; resolve payment exceptions	Organisations and their organisers are recorded nowhere; access to forms, sheets and exports is shared informally.	P6	Organisation register, scoped role grants, account lock and audit log (F01); exception resolution records (F06).	Confirm administration operations and minimum audit scope in PA2.

Proposed solution

UniEvent is a web platform that connects the whole event life cycle. Organisers create an event

and submit it for approval inside the system; Youth Union / Student Association officers approve, reject or request changes. Once an event is published, students reserve tickets, which are held for 15 minutes while they pay with any banking app through a VietQR payment link created by the payOS gateway. payOS notifies UniEvent through a signed webhook, and UniEvent issues a signed QR ticket and e-mails it to the student. Free tickets are issued without payment. At the venue, volunteers scan tickets with a phone browser; used or invalid tickets are rejected instantly, and organisers follow sales and check-in on a real-time dashboard. As an optional (Should-have) extension, check-in and check-out data can be turned into conduct score claims that officers approve in bulk.

Objectives and success criteria

The team's preliminary targets will be refined into functional acceptance criteria (WBS 2.2.1–2.2.3) and measurable non-functional requirements and business rules (WBS 2.2.4) during PA2. Test cases will be designed in PA4 and executed in PA5.

Objective	Preliminary target	How it will be verified
No manual payment reconciliation	For a valid and successfully matched payment, one ticket is created within 60 seconds after receipt of a verified payOS webhook, and e-mail delivery is initiated automatically. Exceptional payments may require manual handling.	E2E tests with a mock adapter in CI plus acceptance evidence from one authorised live payOS channel (Section 3.3).
No overselling, no duplicate tickets	Active reservations and issued tickets must not exceed event capacity; repeated verified payment callbacks must not issue duplicate tickets.	Concurrency and webhook-idempotency cases designed under WBS 4.3.2 and executed under WBS 6.1.1; technical spike under WBS 3.1.4.
Fast and reliable check-in	Preliminary target: at least 95% of scan requests return a valid / used / invalid result within 2 seconds under defined 4G test conditions. Invalid tickets and duplicate check-ins are rejected. Confirm measurement conditions in PA2.	System tests on real Android and iPhone devices.
Transparent approval	Every event shows its approval status, and every decision is kept in a history with the approver's reason.	Acceptance tests for F02–F03.

Objective	Preliminary target	How it will be verified
Real-time visibility	The organiser dashboard reflects recorded ticket sales and check-ins within 5 seconds after server-side confirmation.	Integration and E2E tests for F08.

Operating environment

- ✔ Client: any modern HTML5 browser (Chrome, Edge, Firefox, Safari – latest two versions) on desktop and mobile; the scanner needs a phone with a camera.
- ✔ Server: cloud hosting with HTTPS everywhere (also required by browsers for camera access).
- ✔ External services: payOS payment gateway (VietQR payment links and webhooks) and a transactional e-mail service.

The technology stack and the hardware and software environments are described in Sections 3.2 and 3.3.

Design and implementation constraints

- ✔ Process: waterfall model with V-model test planning; phase gates PA0–PA5 and a final presentation within 11 weeks; implementation in weeks 8–10 (Section 4.1).
- ✔ Team and budget: four students. PA0 set 8–10 h per member per week (about 12 h in W8–W10); on 21/10/2026 the team agreed to 12 h in W4–W5, 10 h in W6–W7, 13 h in W8 and 20 h in W9–W10 (Section 4.3). Only free tiers of hosting and tools are used.
- ✔ Technology: TypeScript for front end and back end, PostgreSQL database, REST API with a WebSocket channel for real-time updates (Section 3.2).
- ✔ Payments: UniEvent does not hold or aggregate funds. The PA5 delivery scope uses one authorised payOS payment channel with direct settlement to its account holder. Multi-organiser onboarding, per-organiser credential management and payment routing

are excluded from PA5. F10 cancellation/refund policy automation remains optional; minimum handling of received-payment exceptions is required under F05–F06.

- ✔ Data protection: minimise collected personal data, apply role-based access controls, and avoid including personal data in QR ticket payloads. Data-handling requirements will be specified in PA2 with reference to applicable Vietnamese legislation, including Law No. 91/2025/QH15.
- ✔ Documentation: course templates for every PA, UML 2.5 for models, the C4 model for architecture, Architecture Decision Records for key decisions.
- ✔ Code quality target: ESLint/Prettier, Conventional Commits and reviewed PRs for repository changes. PA1 sections are co-edited in Google Docs and do not require separate PRs. A future pull_request application build/lint/test CI pipeline and required status checks remain to be configured.

Assumptions and dependencies

- ✔ Live-channel access remains an external dependency. Học coordinates the nominated account holder and TA approval; Minh verifies account eligibility, credentials and signed webhook access in the W4 technical spike (WBS 3.1.4), and Kiệt reviews the spike result. The mock adapter supports deterministic tests and a clearly labelled contingency demo; it does not establish completion of live integration. If authorisation/access is unavailable, Học obtains an explicit revised delivery decision from the TA rather than silently replacing the required channel with a mock.
- ✔ Prospective paid-ticket attendees are assumed to have access to a bank or payment app that supports VietQR; validate accessibility and offer free tickets without requiring such an app.
- ✔ The team plans to seek feedback from Student Volunteer Team organisers and Youth Union officers about event approval, refunds and conduct-score rules during PA2 (WBS

2.1.2); their availability and participation are not yet confirmed.

✔ The detailed problem statement, requirements and business rules will be specified in PA2.

Essential open rules include: payment received after a reservation expires; whether an order is cancellable after payment; matching webhook order code, amount and status; safely retrying email without creating a second ticket; check-in/check-out state transitions; multi-role permissions; and the data-retention/access policy under Law No. 91/2025/QH15. Each rule needs an explicit acceptance test.

2.2 Product Position Statement

Element	Content
For	student clubs and Youth Union / Student Association officers at Vietnamese universities
Who	run events with Google Forms, manual bank-transfer reconciliation and paper attendance lists
The UniEvent	is a web-based event management and ticketing platform
That	integrates event approval, VietQR payment confirmation, signed QR ticket issuance and attendance tracking, with optional conduct-score evidence management
Unlike	disconnected workflows based on Google Forms, spreadsheets and manual bank transfers, or general-purpose event platforms such as Ticketbox and Luma
Our product	focuses on Vietnamese university workflows by integrating institutional event approval, free and low-cost ticketing, automated VietQR payment confirmation and verifiable attendance records in one platform

User-specific position statements (derived from the overall statement; validated in PA2 interviews)

Element	Attendee
For	students who attend campus events
Who	register through forms, pay by bank transfer and wait for a manual confirmation
The UniEvent	attendee portal
That	reserves a ticket, takes payment through VietQR and delivers a signed QR ticket automatically, then shows the attendance result

Element	Attendee
Unlike	Google Forms with manually confirmed transfers
Our product	issues exactly one verified ticket per paid order and keeps a visible attendance record (F05–F07)

Element	Club organiser
For	club organising committees
Who	run approval, registration, payment reconciliation and check-in with disconnected tools
The UniEvent	organiser workspace
That	covers event submission, ticket configuration, check-in staff assignment, automatic payment confirmation and a live dashboard
Unlike	spreadsheets and bank statements, or general platforms such as Ticketbox and Luma that have no institutional approval step
Our product	removes manual reconciliation in the normal case and shows sales and attendance live (F01–F06, F08)

Element	Check-in volunteer
For	volunteers at the venue entrance
Who	look attendees up in printed or shared lists
The UniEvent	mobile-browser scanner
That	validates signed QR tickets for the events they are assigned to
Unlike	list lookup or screenshots that can be reused
Our product	returns a valid, already-used or invalid result (target: 95% of scans within 2 seconds on 4G) and records check-in and check-out atomically (F07)

Element	Approver
For	Youth Union and Student Association officers
Who	receive event plans by e-mail or on paper
The UniEvent	approval queue
That	collects submissions from their unit and records approve, reject-with-reason and request-changes decisions
Unlike	e-mail threads and paper files
Our product	keeps every decision and reason in a searchable history, with optional bulk review of conduct-score evidence (F03, F09)

Element	System administrator
For	the operator responsible for UniEvent access
Who	must control which organisations exist and who holds organiser and approver rights
The UniEvent	administration console
That	manages organisations, scoped role grants, account status and the audit log
Unlike	informal sharing of form, spreadsheet and payment access
Our product	makes every privileged change and data export traceable to a named account (F01)

2.3 Stakeholders and Users

#	Stakeholder / user	Description	Main interests
1	Academic stakeholder – instructor and TA	Teaching staff who supervise the project, review PA submissions and evaluate delivery against course requirements	A complete, consistent process; working software; fair individual contributions

#	Stakeholder / user	Description	Main interests
2	End user – student (attendee)	Any student who joins events	Convenient ticket purchasing, access to issued tickets, accurate attendance records and optional conduct-score history if F09 is implemented
3	End user – club organiser	Member of a club's organising committee	No manual reconciliation, real-time sales and attendance, easy approval process
4	Potential domain expert / pilot organisation – Student Volunteer Team	A student volunteer organisation known to the team lead; workflow validation and possible pilot participation are subject to the organisation's agreement	A workflow that fits real volunteer activities; reliable attendance records; a chance to pilot the system
5	End user – check-in volunteer	Club member working at the venue entrance	Fast and unambiguous scanning, works on their own phone
6	End user – Youth Union / Student Association officer	Approves events and (optionally) conduct score claims	Clear event information, decision history, reliable attendance evidence, bulk review
7	End user – system administrator	Operator who registers organisations, grants organiser and approver roles scoped to an organisation, and records the resolution of payment exceptions	A complete organisation register, secure scoped role assignments and traceable changes
8	External system – payOS	VietQR payment gateway used through the single authorised PA5 payment channel	Correct use of its API and webhook, secure handling of keys
9	External system – e-mail service	Transactional e-mail provider that delivers tickets and notifications	Correct use of its API, reliable delivery, secure handling of keys
10	Development team	Four students building UniEvent	Achievable scope, clear requirements, balanced workload, portfolio value

User roles on the system (one account may hold several roles; organiser, approver and checker rights are scoped)

Role	Who holds it	Granted by	What the role does in UniEvent	Features
Guest	Any visitor without signing in	—	Browses published event pages	Public pages
Attendee	Any student with a verified university e-mail	Self-registration	Reserves and pays for tickets, holds QR tickets, views own attendance	F01, F05–F07; F09–F10 if built
Organiser	Member of a club organising committee	Administrator, for one organisation	Creates and submits events, configures ticket types, assigns check-in staff, monitors the dashboard, exports attendees	F01–F06, F08; F10 if built
Checker	Volunteer working at one event	Organiser, for one event	Scans tickets and records check-in and check-out	F07
Approver	Youth Union / Student Association officer	Administrator, for one approving unit	Approves, rejects or requests changes to proposals from their unit; optional bulk review of conduct-score evidence	F03; F09 if built
Administrator	System operator appointed by the team for PA5	Seeded at deployment; further admins by an existing admin	Manages organisations, role grants, account status, the audit log and payment-exception records	F01, F06 exceptions

Actor × feature responsibility matrix (preliminary; authority is restricted by event/organisation scope)

C = create/change; V = authorised view; X = primary execution/use; — = no default permission. Organiser, checker and approver rights apply only within their assigned organisation, event or unit.

Feature	Attendee	Organiser	Checker	Approver	Admin
F01 Account and sign-in	X own account	X own account	X own account	X own account	X own account; C lock/unlock accounts
F01 Organisations and role grants	—	C checker assignment for own events	—	—	C organisations; C organiser and approver

Feature	Attendee	Organiser	Checker	Approver	Admin
					grants; V audit log
F02 Event proposal	V published events	C draft, submit, revise	—	V proposals of own unit	V audit
F03 Event decision	V published status	V status and reasons	—	C approve / reject / request changes	V audit
F04 Ticket configuration	V offered ticket types	C ticket types and limits	—	V in proposal review	V audit
F05 Reserve / pay	C own reservation and payment	V orders of own events	—	—	V exception audit
F06 Issue ticket / exceptions	X receives ticket	V order status and exceptions of own events	—	—	C exception resolution record
F07 Check-in / check-out	X presents ticket	V attendance of own events	C check-in and check-out on assigned events	V attendance if F09 is used	V audit
F08 Dashboard / CSV	—	X dashboard; C CSV export	—	—	V operational audit
F09 Conduct-score evidence (Should)	V own records	V drafts of own events	—	C bulk review	V audit
F10 Cancellation / refund (Should)	C cancellation request	C refund decision under the policy	—	—	V audit

Key data: source, stewardship and consumers (conceptual mapping, not a database schema)

Data / future entity	Created by	Updated by	Used / exported by	PA2–PA3 decision
Account	Student self-registration with a university e-mail	Account holder (profile); administrator (lock/unlock)	All signed-in roles	University e-mail policy and retention period.
Organisation (club, Youth Union or Student Association unit)	Administrator	Administrator (edit, deactivate)	Organisers and approvers within scope; attendees see the organiser name	Unit hierarchy and how a club proves its status.
Role grant	Administrator (organiser, approver); organiser (checker for own event)	The granting role revokes; an administrator may revoke any grant	Access checks on every request; audit log	Grant request flow, expiry and minimum audit fields.
Event / approval decision	Organiser (event form); approver (decision)	Organiser edits draft; approver changes decision status	Attendees view published events; organisers and officers audit	State transitions, change requests and ownership.
Ticket type / inventory	Organiser (price, sale window, cap, per-account limit)	Organiser before sale; system reserves and releases atomically	Attendees choose; organisers report	Transaction boundaries and overbooking constraints.
Reservation / payment status	Attendee reservation; verified payOS webhook	System only	Attendee receipt; organiser dashboard	Signature, amount, order code, expiry, idempotency.
Payment exception record	System flags a paid-but-unissued order	Administrator records the resolution; organiser	Administrator, organiser of the event	Closure evidence and notification rules.

Data / future entity	Created by	Updated by	Used / exported by	PA2–PA3 decision
		confirms payer notification		
Ticket / attendance	System issues QR; checker scans	Server accepts valid state transitions atomically	Attendee proof; organiser CSV; optional officer review	Retention and check-in/check-out dispute procedure.
Audit log	System records grants, exports and exception resolutions	Append-only; nobody edits	Administrator	Retention under Law No. 91/2025/QH15.

2.4 Scope and Key Features

Features are prioritised using MoSCoW. Must-have features (F01–F08) define the required PA5 delivery scope. Should-have features (F09–F10) are conditional extensions and must not delay Must-have delivery. Could-have features (F11–F14) form a future backlog, excluded from the PA5 implementation baseline. Won't-have features (F15–F16) are outside the current project scope.

ID	Feature	Primary actor(s)	Description	Priority (M/S/C/W)
F01	Accounts, organisations and role administration	All roles; Administrator; Organiser	Sign-up/sign-in with a university e-mail address; one account may hold several roles. Administrators register and deactivate organisations, grant and revoke organiser and approver roles scoped to an organisation, lock or unlock accounts and view the audit log. Organisers assign check-in staff to their own events. Guests browse published events without an account.	Must

ID	Feature	Primary actor(s)	Description	Priority (M/S/C/W)
F02	Event creation and submission	Organiser	Organisers create an event (time, venue, capacity, poster, description) and submit it to their approving unit.	Must
F03	Event approval workflow	Approver	Approvers approve, reject with a reason, or request changes; every decision is kept in an approval history.	Must
F04	Ticket type configuration	Organiser	Price (0 VND for free tickets), quantity, sales window and per-account purchase limit for each ticket type. Audience restrictions are part of F11.	Must
F05	Ticket reservation and VietQR payment	Attendee	A reservation holds tickets for 15 minutes. Paid tickets use the single authorised PA5 payOS channel; free tickets skip payment. Multi-organiser channel management is excluded.	Must
F06	Payment confirmation and ticket issuing	System; Attendee; Administrator	Verifies the payOS callback signature, order, amount and status; processes callbacks idempotently; issues and e-mails one signed QR ticket. Paid-but-unissued orders are flagged and resolved by an administrator.	Must
F07	QR check-in and check-out	Checker	Mobile-browser scanner validates signed tickets for assigned events, records check-in and check-out, and rejects invalid tickets, duplicate check-ins and duplicate check-outs.	Must
F08	Real-time organiser dashboard	Organiser	Tickets sold, revenue and check-in rate updated in real time; CSV export of attendees.	Must
F09	Conduct score records	Approver; Attendee	Optional: generates draft conduct-score evidence from check-in/check-out data with basic bulk review by authorised officers.	Should

ID	Feature	Primary actor(s)	Description	Priority (M/S/C/W)
			Appeals and integration with official university systems are excluded from PA5.	
F10	Cancellation and refund policy	Attendee; Organiser	Ticket cancellation and refund decisions under the refund policy; refunds are recorded, money is returned manually.	Should
F11	Discount codes and special tickets	Organiser; Attendee	Early-bird tickets, club-member-only tickets (requires an organiser-maintained member list) and conditional discount codes.	Could
F12	Waitlist	Attendee; Organiser	Students join a waitlist for sold-out ticket types and are invited in order when tickets are released.	Could
F13	Offline check-in	Checker	Progressive web app mode that checks tickets without network access and synchronises later.	Could
F14	AI writing assistant for organisers	Organiser	Suggests event descriptions and summarises post-event feedback; the organiser approves every suggestion.	Could
F15	Seat maps and reserved seating	Attendee	Choosing a specific seat when buying a ticket.	Won't
F16	Native mobile applications	All roles	iOS/Android apps; UniEvent stays a responsive web app.	Won't

Shaded rows are Must-have features.

PA5 payment delivery and exception handling

The required delivery is one authorised live payOS payment channel supporting F05–F06: reservation, payment-link creation, signed callback verification, order/amount/status matching, idempotent ticket issuance and ticket delivery. Free tickets bypass payment. CI uses the mock adapter; acceptance includes evidence from the authorised live channel. The account holder and permission to test remain to be confirmed by Học with the TA by W4 Thursday, before real transfers.

If a received payment cannot produce a valid ticket, UniEvent retains transaction evidence, flags the exception for authorised review and prevents duplicate processing. Kiệt coordinates technical investigation; Học contacts the organiser/account holder. The account holder handles payer communication and any manual return of funds outside UniEvent. Resolution and payer-notification evidence are recorded before closure. This minimum workflow is required even when optional F10 is deferred; no automated refund or custody of funds is promised.

Out of scope:

- ✔ Holding or aggregating organisers' funds, distributing payouts, and onboarding or routing payments across multiple organiser-owned payment channels are outside PA5. The delivery baseline contains one authorised payOS channel.
- ✔ International card payments and currencies other than VND.
- ✔ Native iOS/Android applications – UniEvent is a responsive web application only.
- ✔ Direct integration with the university's official conduct score or student-information systems (UniEvent exports records instead).
- ✔ Seat maps and reserved seating.

Changes since PA0

PA0 is the baseline for this document. PA1 retains F01–F10 priority categories while clarifying QR check-in/check-out rules and the optional scope of conduct-score features. F11–F14 are recorded as Could-have backlog; F15–F16 are Won't-have. Other material changes are listed below.

Item	PA0	PA1	Section	Changed by
Feature list	F01–F10 (eight Must, two Should)	F11–F14 added as Could-have backlog; F15–F16 recorded as Won't-have. F01–F10 keep their PA0 priority categories.	2.4	Thái Quang Huy

Item	PA0	PA1	Section	Changed by
F01 administration	Administrators “manage clubs, accounts, roles and system-wide policies”	Administrators manage organisations, organiser/approver grants scoped to an organisation, account status and the audit log; organisers assign check-in staff to their own events (new WBS 5.2.13).	2.3, 2.4	Thái Quang Huy
F04 Ticket types	Price, quantity, sales window, per-person limit	Per-account purchase limit; audience restrictions such as club-member-only tickets moved to F11.	2.4	Thái Quang Huy
F07 Attendance	QR check-in; reject invalid/used tickets	Separate check-in and check-out with valid state transitions and duplicate-scan rejection.	2.4	Thái Quang Huy
F09 Conduct score	Records and bulk review	Remains Should-have; basic draft evidence records and bulk review only; appeals and official-system integration excluded from PA5.	2.4	Thái Quang Huy
Stakeholders and users	Five target-user groups	Ten stakeholders and six system roles, including Guest, the instructor/TA, the pilot organisation and external systems.	2.3	Thái Quang Huy
Payment scope	Payments go to each organiser's own payOS account	One authorised live payOS channel for PA5; multi-organiser onboarding and routing excluded.	2.4	Lê Nguyễn Quốc Học
Payment test environment	Mock adapter mentioned as a risk mitigation	Mock adapter for CI and labelled contingency demo; live acceptance evidence from the authorised channel.	3.3	Lê Nguyễn Quốc Học
Technology	“Not final – confirmed in PA1”	Selected: Next.js, NestJS, PostgreSQL/Prisma; reasons in Section 3.2. Versions and hosting providers confirmed during setup.	3.2	Trần Vũ Anh Kiệt

Item	PA0	PA1	Section	Changed by
Effort estimate	About 43 h for Must-have implementation, plus tests/review	94 h for WBS 5.2; 120 h including foundation and unit-test/CI work; 427 h for the required project.	4.3	Lê Nguyễn Quốc Học
Capacity	8–10 h/member/week; about 12 h in W8–W10	Raised by all-member agreement on 21/10/2026: 12 h in W4–W5, 13 h in W8, 20 h in W9–W10; demand is covered with 8 h slack in each window (Section 4.3).	4.3–4.5	Lê Nguyễn Quốc Học

3 Proposed Solution

3.1 Preliminary Architecture

UniEvent is used by five authenticated roles – attendee, organiser, checker, approver and administrator – and by unauthenticated guests who browse published events; it depends on two external systems: payOS, whose API issues VietQR payment links and notifies UniEvent of completed payments through webhooks, and an e-mail service, which delivers tickets and notifications

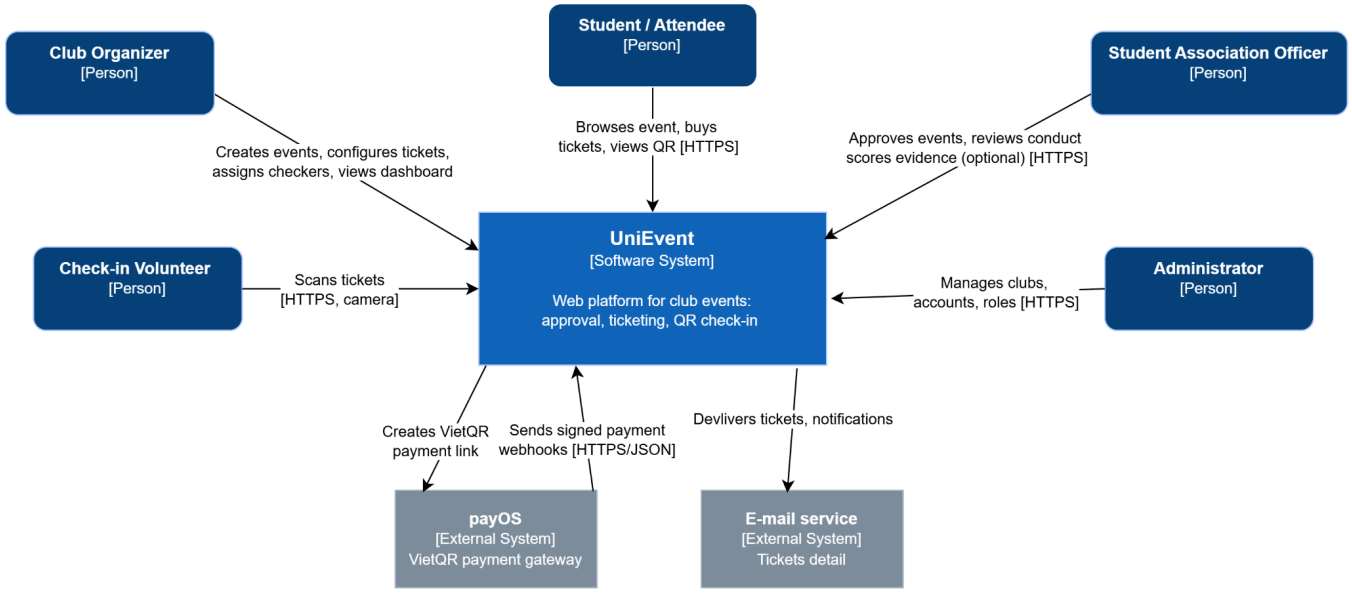


Figure 1. C4 System Context diagram of UniEvent

3.2 Technology Stack

Layer	Technology	Reason for choice
Front end	Next.js (React, TypeScript), Tailwind CSS	Responsive pages and server-side rendering; large ecosystem; TypeScript types shared with the back end

Layer	Technology	Reason for choice
Back end	NestJS on Node.js LTS (TypeScript), REST + WebSocket (Socket.IO)	Module structure matches our feature modules; dependency injection makes unit testing easy; built-in WebSocket gateway for the real-time dashboard
Database	PostgreSQL with Prisma ORM	ACID transactions and row-level locks prevent overselling; relational model fits events, tickets and orders; versioned migrations
Payment and e-mail	payOS (VietQR) behind a PaymentProvider interface with a mock adapter; transactional e-mail service (e.g., Resend)	VietQR is the payment method students already use; the mock adapter keeps CI and demos independent of the real gateway
Hosting / deployment	Docker; Vercel (web), Render or Railway (API), Neon (PostgreSQL)	Candidate providers; verify quotas, supported connections and HTTPS before choosing the deployment configuration.
Testing & CI	Jest, Supertest, Playwright, k6, GitHub Actions, ESLint, SonarQube Cloud	Covers unit, integration, end-to-end and load testing; CI runs on every pull request; static analysis as a quality gate

The PA1 technology choices in Section 3.2 are selected for this project. Exact library versions are pinned at implementation setup; upgrades require compatibility checks. Hosting and e-mail providers remain deployment choices to be verified against quotas, WebSocket support and the approved budget.

3.3 Hardware and Operating Environment

Environment	Hardware / software
Development	Members' laptops (Windows, macOS or Linux, 8 GB RAM or more) Node.js LTS, Docker Desktop, Git, VS Code with ESLint/Prettier, Figma, draw.io / PlantUML
Testing	Desktop browsers: Chrome, Edge, Firefox (latest) At least one Android phone (Chrome) and one iPhone (Safari) for QR scanning and responsive tests 4G connection for check-in performance tests; GitHub Actions runners for automated tests

Environment	Hardware / software
Staging / production	Web app, API container and PostgreSQL on providers selected against deployment requirements and available quotas. One nominated, authorised payOS channel is required for live acceptance testing; account-holder and TA permission must be recorded before real transfers.
End users	Any device with a modern HTML5 browser; a camera for the check-in scanner; a banking app for VietQR payments

4 Project Plan

4.1 Process and Phase Gates

The project follows the waterfall model. Each phase ends with a phase gate (a PA submission). After a PA is submitted it becomes a baseline; later changes must be updated in the document and listed in the next PA.

Course weeks map to calendar dates as follows; W1 starts on Monday 5 October 2026. A deadline published on Moodle overrides the planned week-end marker, and each internal freeze moves with it (drafts at T−2 days, reviewed freeze at T−1 day).

Week	Dates (2026)	Gate or fixed date
W1	05/10 – 11/10	—
W2	12/10 – 18/10	PA0 submitted Wed 14/10, 22:00
W3	19/10 – 25/10	PA1 due Wed 21/10, 22:00
W4	26/10 – 01/11	—
W5	02/11 – 08/11	PA2
W6	09/11 – 15/11	—
W7	16/11 – 22/11	PA3
W8	23/11 – 29/11	PA4
W9	30/11 – 06/12	—
W10	07/12 – 13/12	PA5
W11	14/12 – 20/12	Final presentation and demo

PA	Phase	Due	Exit criteria
PA1	Initiation & planning	End of week 3	Vision, scope, WBS, estimates, schedule and risk register reviewed by all members; repository, board and Discord set up; PDF and source submitted and committed to /pa/PA1
PA2	Requirements	End of week 5	Every Must-have FR numbered with acceptance criteria; measurable NFRs; business rules; all Must-have use cases specified; mockups for main use cases; traceability F to FR to UC complete; review record with all issues closed
PA3	Design	End of week 7	C4 context and container, component and deployment diagrams completed; at least 3 ADRs; consistent class, sequence and state diagrams; normalized database schema with ERD, table specifications and indexes; screen flow and accessibility checklist; all Must-have FRs traced to design elements; design consistency reviewed.
PA4	Test planning	End of week 8	Test plan completed with scope, levels, types, environment, entry/exit criteria and defect management; EP, BVA and decision tables applied; test cases documented with preconditions, test data, steps and expected results; every Must-have FR covered by at least one test case.
PA5	Implementation, testing & deployment	End of week 10	All Must-have features implemented and test cases executed with results recorded; no open critical or major defects; unit-test coverage reported; at least 2 use cases × 2 scenarios automated in CI; code reviewed through PRs and static analysis completed; application deployed with demo video; retrospective completed.
Final	Presentation & demo	Week 11	Slides cover all required content and are rehearsed with a timer; every member speaks at least 2 minutes in English; key scenarios demonstrated live with a backup video available.

4.2 Work Breakdown Structure (WBS)

The deliverable-oriented WBS contains Level 0 (UniEvent project), seven Level-1 workstreams, 26 Level-2 deliverables and 72 Level-3 work packages. Figure 2 shows Levels 0–2; Table 1 maps

each deliverable to its Level-3 IDs. Section 4.3 lists every work package with its owner, week, estimate and output. Each Level-3 work package is executed by exactly one owner; a different member reviews it, and the reviewer is recorded on the GitHub issue.

Work packages will be tracked on GitHub Projects with the same WBS ID, owner, estimate, due date, evidence, and status. Monitoring (1.3) is recurring from W2 to W10. Work package 3.1.4 is an exploratory payment/concurrency technical spike scheduled in W4 to reduce feasibility uncertainty; the remaining architecture design activities (3.1.1–3.1.3) are scheduled in W6–W7 after the PA2 gate. The W4 spike does not constitute approval of the PA3 design baseline. Should-have features (F09/F10) remain conditional and are excluded from the committed PA5 delivery baseline.

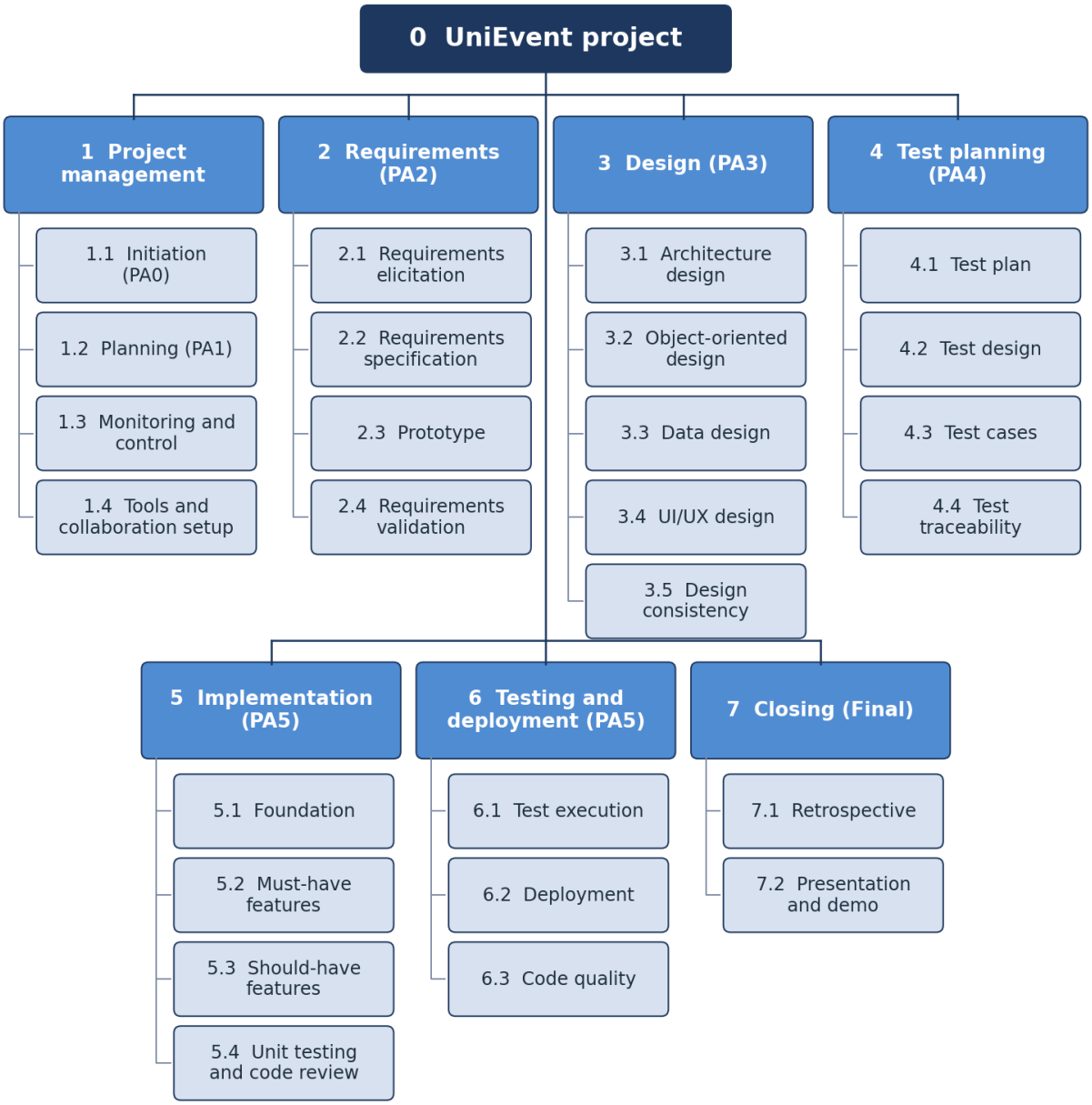


Figure 2. WBS of UniEvent (levels 1-2)

Table 1. Deliverables (level 2) and their work packages (level 3)

WBS ID	Phase / deliverable	Work packages	Owners / phase gate
1	Project management		
1.1	Initiation (PA0)	1.1.1–1.1.2 (2)	A, B / PA0
1.2	Planning (PA1)	1.2.1–1.2.5 (5)	A, B, C, D / PA1
1.3	Monitoring and control	1.3.1–1.3.3 (3)	A, B / PA5
1.4	Tools and collaboration setup	1.4.1–1.4.3 (3)	B / PA1
2	Requirements (PA2)		
2.1	Requirements elicitation	2.1.1–2.1.2 (2)	B / PA2
2.2	Requirements specification	2.2.1–2.2.8 (8)	A, B, C, D / PA2
2.3	Prototype	2.3.1 (1)	C / PA2
2.4	Requirements validation	2.4.1–2.4.2 (2)	B, C / PA2
3	Design (PA3)		
3.1	Architecture design	3.1.1–3.1.4 (4)	C, D / PA3
3.2	Object-oriented design	3.2.1–3.2.3 (3)	A, B / PA3
3.3	Data design	3.3.1–3.3.2 (2)	D / PA3
3.4	UI/UX design	3.4.1 (1)	C / PA3
3.5	Design consistency	3.5.1 (1)	A / PA3
4	Test planning (PA4)		
4.1	Test plan	4.1.1 (1)	D / PA4
4.2	Test design	4.2.1–4.2.2 (2)	A, B / PA4
4.3	Test cases	4.3.1–4.3.3 (3)	A, C, D / PA4
4.4	Test traceability	4.4.1 (1)	B / PA4
5	Implementation (PA5)		
5.1	Foundation	5.1.1–5.1.2 (2)	B, C / PA5
5.2	Must-have features	5.2.1–5.2.13 (13)	A, B, C, D / PA5

WBS ID	Phase / deliverable	Work packages	Owners / phase gate
5.3	Should-have features	5.3.1–5.3.2 (2)	B, D / PA5
5.4	Unit testing and code review	5.4.1–5.4.2 (2)	C, D / PA5
6	Testing and deployment (PA5)		
6.1	Test execution	6.1.1–6.1.3 (3)	B, D / PA5
6.2	Deployment	6.2.1–6.2.2 (2)	A, D / PA5
6.3	Code quality	6.3.1 (1)	A / PA5
7	Closing (Final)		
7.1	Retrospective	7.1.1 (1)	A / PA5
7.2	Presentation and demo	7.2.1–7.2.2 (2)	A, C / Final

4.3 Estimation

Effort is estimated in person-hours (one person working for one hour), not elapsed calendar time. Each estimate covers the owner's work to produce the stated output, including addressing review comments. The team reviewed these estimates at the PA1 freeze; actual hours will be logged on each GitHub issue and compared in WBS 7.1.1. Work packages are generally scoped to approximately 3–9 person-hours; actual work exceeding the estimate must be re-estimated or split.

Three estimation methods are used: EJ = expert judgement (the assigned team member assesses the steps and complexity); AN = analogy (reference to comparable tasks or reusable templates/components); and 3P = three-point estimation, with optimistic O, most-likely M, and pessimistic P hours. For 3P packages, the weighted estimate is (O + 4M + P) / 6. For example, the payment-webhook technical spike uses 4/8/12 hours, yielding 8 hours.

The estimates assume a responsive web application, an existing GitHub/Discord setup, reusable web libraries, payOS sandbox access, a transactional email provider, and no native mobile app or official university information-system integration. They are sensitive to external-service constraints, especially payOS webhooks and concurrency, and must be revisited after the

week-4 spike. Estimates for optional F09 and F10 are shown separately and excluded from the required PA5 effort.

Table 2.Table 2. Work packages (level 3) with owners, weeks, estimates and outputs

WBS ID	Work package	Owner	Week(s)	Effort (h)	Method / assumptions	Output (done when)
1	**Project Management**			**74**	Phase subtotal	
1.1.1	Finalize project idea and feasibility assessment	A	W1–W2	5	EJ; PA0 idea and competitor scan	Project idea section of T07-PA0 approved for PA0 submission
1.1.2	Finalize team contract and responsibilities	B	W1–W2	4	EJ; four-member contract review	Signed team contract in T07-PA0
1.2.1	Define project vision, scope and stakeholders	B	W2–W3	6	EJ; reuse approved PA0 scope	PA1 Section 2 reviewed by Học
1.2.2	Prepare preliminary architecture and technology choices	C	W2–W3	6	EJ; preliminary C4 and stack only	PA1 Section 3 with Figure 1, reviewed by Minh
1.2.3	Create WBS and effort estimation	A	W2–W3	8	EJ; 72 WPs and cross-check	PA1 Sections 4.2–4.3 with Figure 2, reviewed by Kiệt
1.2.4	Prepare schedule, roles and cost estimate	A	W2–W3	8	EJ; Gantt and cost workbook	PA1 Sections 4.4–4.6 with Figure 3, reviewed by Kiệt

WBS ID	Work package	Owner	Week(s)	Effort (h)	Method / assumptions	Output (done when)
1.2.5	Prepare risk register and tools & collaboration section	D	W2–W3	6	EJ; at least five scored risks	PA1 Sections 5–6, reviewed by Huy
1.3.1	Prepare weekly status reports and meeting records	A	W2–W10	9	AN; nine weekly reports and minutes	W2–W10 status reports in /docs/management/weekly by Sunday 22:00
1.3.2	Monitor task-board progress and commitments	A	W2–W10	7	AN; weekly board checks	Board reviewed weekly; overdue items listed in the weekly report
1.3.3	Track risks, issues and approved changes	B	W2–W10	6	EJ; risks and changes logged weekly	Risk and change log updated in each weekly report
1.4.1	Configure Git repository and branch protection	B	W2	4	AN; existing GitHub repository	Repository structure and active “Protect main” ruleset
1.4.2	Set up GitHub Projects task board	B	W2	3	AN; one GitHub Projects board	“Team Management” board with issues, labels and status options
1.4.3	Set up team communication channel	B	W2	2	AN; existing Discord server	Discord server with issue notifications configured
2	**Requirements (PA2)**			**75**	Phase subtotal	

WBS ID	Work package	Owner	Week(s)	Effort (h)	Method / assumptions	Output (done when)
2.1.1	Collect stakeholder needs and pain points	B	W4	6	EJ; short stakeholder interviews	Interview notes in /docs/requirements
2.1.2	Validate event workflows and project assumptions	B	W4–W5	5	EJ; validate with other organisers	Validated assumption list with sources
2.2.1	Specify numbered FRs and acceptance criteria for F01–F03	A	W4–W5	6	AN; three features, shared template	Numbered FRs with acceptance criteria for F01–F03
2.2.2	Specify numbered FRs and acceptance criteria for F04–F06	A	W4–W5	7	AN; three ticket/payment features	Numbered FRs with acceptance criteria for F04–F06
2.2.3	Specify numbered FRs and acceptance criteria for F07–F08	C	W4–W5	6	AN; two check-in/dashboard features	Numbered FRs with acceptance criteria for F07–F08
2.2.4	Specify measurable NFRs and business rules	D	W4–W5	7	EJ; measurable targets and rules	NFR list with targets and business-rule table
2.2.5	Develop use-case model and actor relationships	B	W4–W5	5	AN; actors and UML relationships	UML use-case diagram with actor generalisation
2.2.6	Specify account, administration, event and	D	W5	7	AN; concise numbered UC flows	UC specifications for F01–F03, including

WBS ID	Work package	Owner	Week(s)	Effort (h)	Method / assumptions	Output (done when)
	approval use cases					administrator use cases
2.2.7	Specify ticketing and payment use cases	A	W5	5	EJ; multiple payment-related UC flows	UC specifications for F04–F06
2.2.8	Specify check-in and dashboard use cases	C	W4–W5	4	AN; split from former 2.2.7	UC specifications for F07–F08
2.3.1	Create mockups for main use cases	C	W4–W5	8	AN; low-fidelity key-use-case mockups	Mockups for every main use case, linked to UC IDs
2.4.1	Build feature–FR–UC traceability matrix	C	W5	4	AN; F–FR–UC links in matrix	Traceability matrix F → FR → UC
2.4.2	Conduct requirements review and resolve blocking issues	B	W5	5	EJ; one review + issue resolution	Review record with all blocking issues closed
3	**Design (PA3)**			**68**	Phase subtotal	
3.1.1	Create C4 context and container diagrams	C	W6	5	AN; C4 context/container templates	C4 context and container diagrams
3.1.2	Create component and deployment diagrams	C	W6–W7	6	EJ; component/deployment scope	Component and deployment diagrams

WBS ID	Work package	Owner	Week(s)	Effort (h)	Method / assumptions	Output (done when)
3.1.3	Document at least three architecture decision records (ADRs)	D	W6–W7	6	AN; three concise ADRs	At least three ADRs in /docs/design
3.1.4	Run payment-webhook and concurrent-reservation technical spike	D	W4	8	3P (4/8/12); authorised channel and two technical risks	Spike report: payOS channel access, webhook signature, idempotency and concurrency results
3.2.1	Prepare class diagrams and class specifications	B	W6–W7	7	EJ; domain classes from UCs, with specs	Class diagram and class specifications traced to UCs
3.2.2	Prepare key sequence diagrams	A	W6–W7	6	AN; two to three key scenarios	Two to three sequence diagrams
3.2.3	Prepare key state diagrams	B	W6–W7	5	AN; ticket/order /event states	State diagrams for ticket, order and event
3.3.1	Create ERD and database table specifications	D	W6–W7	8	EJ; core entities and table specs	ERD and table specifications
3.3.2	Review normalization, constraints and indexes	D	W7	5	EJ; review keys/index trade-offs	Normalisation, constraint and index notes
3.4.1	Define screen flow, key screens and accessibility checklist	C	W6–W7	7	AN; reuse PA2 mockups	Screen flow, key screens and accessibility checklist

WBS ID	Work package	Owner	Week(s)	Effort (h)	Method / assumptions	Output (done when)
3.5.1	Review design consistency and FR-to-design traceability	A	W7	5	EJ; all Must-have FR links	Consistency review record; every Must-have FR traced to design
4	**Test Planning (PA4)**			**36**	Phase subtotal	
4.1.1	Define test scope, levels, types, environment, entry/exit criteria and defect workflow	D	W8	6	EJ; complete test-plan template	Test plan document
4.2.1	Design equivalence-partitioning (EP) and boundary-value (BVA) scenarios	A	W8	4	AN; selected boundaries/ partitions	EP and BVA tables
4.2.2	Design decision-table scenarios for business rules	B	W8	4	AN; key conditional business rules	Decision tables for key business rules
4.3.1	Write account and event-approval test cases	A	W8	5	AN; reusable test-case template	Test cases for F01–F03
4.3.2	Write ticketing and payment test cases	D	W8	7	EJ; payment negative-path coverage	Test cases for F04–F06, including negative paths
4.3.3	Write QR check-in and dashboard test cases	C	W8	6	AN; QR/dashboard test coverage	Test cases for F07–F08

WBS ID	Work package	Owner	Week(s)	Effort (h)	Method / assumptions	Output (done when)
4.4.1	Build and review FR-to-test-case coverage matrix	B	W8	4	AN; test cases linked to all Must FRs	Coverage matrix: every Must-have FR has at least one test case
5	**Implementation (PA5)**			**136**	Phase subtotal	
5.1.1	Set up application skeleton and database foundation	C	W8	6	EJ; established web stack; seed DB	Running Next.js/NestJS skeleton with migrations and seed data
5.1.2	Set up initial CI and shared configuration	B	W8	4	AN; basic PR build/test workflow	PR build/lint/test workflow set as a required status check
5.2.1	Implement university-email authentication and role-based access (F01)	B	W9	8	EJ; library auth; no campus SSO	Merged PR: sign-up/sign-in and role checks with unit tests
5.2.2	Implement event creation and submission (F02)	A	W9	6	AN; ordinary event CRUD/forms	Merged PR: event form and submission with unit tests
5.2.3	Implement event approval and decision history (F03)	A	W9	7	AN; status workflow and history	Merged PR: approval queue and decision history with unit tests
5.2.4	Implement ticket types, inventory and sales-window rules (F04)	C	W9	6	AN; core ticket CRUD/const raints	Merged PR: ticket-type rules with unit tests

WBS ID	Work package	Owner	Week(s)	Effort (h)	Method / assumptions	Output (done when)
5.2.5	Implement ticket reservation, concurrency safeguards and 15-minute expiry (F05)	C	W9	9	3P (5/9/13); reuse tested DB transactions	Merged PR: reservation, hold and expiry with concurrency tests
5.2.6	Integrate payOS checkout and VietQR payment initiation (F05)	D	W9	8	3P (4/8/12); authorised payOS channel and mock fixtures	Merged PR: payment-link creation through the PaymentProvider interface
5.2.7	Implement verified payment webhooks and duplicate-event handling (F06)	D	W9	9	3P (5/9/13); webhook/idempotency	Merged PR: signed, idempotent webhook handling with tests
5.2.8	Implement QR ticket issuing and email delivery (F06)	A	W9–W10	8	EJ; mail provider and QR lib	Merged PR: one signed QR ticket per paid order, e-mailed
5.2.9	Implement mobile-browser QR scanning interface (F07)	C	W9–W10	7	AN; mobile camera/QR library	Merged PR: scanner page working on Android and iPhone
5.2.10	Implement atomic QR check-in and validated check-out (F07)	C	W10	9	3P (5/9/13); atomic check-in, stateful check-out	Merged PR: check-in/check-out state transitions with repeated-scan tests

WBS ID	Work package	Owner	Week(s)	Effort (h)	Method / assumptions	Output (done when)
5.2.11	Implement real-time organiser dashboard (F08)	B	W9–W10	8	EJ; lightweight live updates only	Merged PR: live dashboard over WebSocket
5.2.12	Implement attendee-list and statistics CSV export (F08)	B	W10	4	AN; simple CSV export	Merged PR: CSV export limited to authorised organisers
5.2.13	Implement organisation management and scoped role assignment (F01)	B	W9	5	AN; reuses 5.2.1 guards and 5.2.2 form components	Merged PR: admin organisation CRUD, organiser/approver grants, checker assignment, audit entries
5.3.1	Implement optional attendance/conduct-score records (F09)	B	Conditional	7	EJ; optional basic records only	Only if R8 allows: draft evidence records and bulk review
5.3.2	Implement optional ticket cancellation and refund handling (F10)	D	Conditional	9	EJ; optional policy workflow only	Only if R8 allows: cancellation requests and recorded refunds
5.4.1	Consolidate module unit tests and coverage report	C	W10	8	EJ; tests are written inside each feature WP	Coverage report and list of untested modules
5.4.2	Automate two use cases × two	D	W10	8	EJ; 2 UCs × 2 scenarios in CI	Four automated E2E scenarios

WBS ID	Work package	Owner	Week(s)	Effort (h)	Method / assumptions	Output (done when)
	scenarios and PR checks in CI					running on every PR
6	**Testing and Deployment (PA5)**			**36**	Phase subtotal	
6.1.1	Execute Must-have test cases and record results	D	W10	8	EJ; run all Must-have test cases	Executed test cases with pass/fail results
6.1.2	Record defects, verify fixes and run regression tests	B	W10	7	EJ; one fix/regression cycle	Defect log and regression results
6.1.3	Prepare test-exit and defect summary report	B	W10	5	AN; results and exit template	Test report against the exit criteria
6.2.1	Deploy application and verify environment configuration	D	W9–W10	7	EJ; free-tier hosting where viable	Deployed staging/production URL with verified secrets and HTTPS
6.2.2	Run deployment smoke tests and record demo video	A	W10	4	EJ; end-to-end smoke/video	Smoke-test record and demo video
6.3.1	Run static analysis and final code-quality review	A	W10	5	AN; ESLint/static check/report	Static-analysis report with critical findings closed
7	**Closing (PA5–Final)**			**18**	Phase subtotal	
7.1.1	Prepare retrospective and	A	W10	5	EJ; plan-vs-actu	Retrospective with

WBS ID	Work package	Owner	Week(s)	Effort (h)	Method / assumptions	Output (done when)
	plan-versus-actual analysis				al and AI lessons	plan-versus-actual hours
7.2.1	Prepare final presentation slides	A	W11	7	AN; standard final slide deck	Final slide deck
7.2.2	Prepare demo scenario and coordinate timed rehearsal	C	W11	6	EJ; 4 speakers, timed rehearsal	Demo script and timed rehearsal record

Table 3. Estimated effort summary

WBS group	Effort	Notes
1. Project Management	74	Continuous tasks across weeks 2–10
2. Requirements (PA2)	75	
3. Design (PA3)	68	
4. Test Planning (PA4)	36	
5. Implementation (PA5)	136	Includes optional F09/F10: 16 h
6. Testing and Deployment (PA5)	36	
7. Closing (PA5–Final)	18	PA5 retrospective and Final
TOTAL INCLUDING OPTIONAL	443	Required scope: 427

Scope totals. Total including optional F09/F10 = 443 person-hours; required scope excluding F09/F10 (16 h) = 427 person-hours.

Responsibility versus effort. Owner totals including optional work are A = 117 h, B = 114 h, C = 103 h, D = 109 h; for the required scope they are A = 117 h, B = 107 h, C = 103 h, D = 100 h. Each total is

the planned execution time of that member, because every work package has a single executing owner. Actual hours are logged on GitHub issues.

Effort reconciliation. Table 2 contains 72 work packages totalling 443 person-hours with F09/F10 included. The required-scope figure is 427 person-hours after excluding optional WBS 5.3.1 (7 h) and 5.3.2 (9 h). WBS 2.2 sums to 47 h and WBS 5.2 sums to 94 h. These are effort figures, not durations; Table 2 is the source of truth for every later total.

Capacity reconciliation. W8–W10 require 204 person-hours: 36 h for PA4, 120 h for required implementation and unit-test/CI work, 36 h for testing/deployment, 5 h for the retrospective and 7 h for monitoring. The availability agreed on 21/10/2026 provides 4 × 13 + 4 × 20 × 2 = 212 h in W8–W10, leaving 8 h of slack, and 4 × 12 × 2 = 96 h in W4–W5 for 88 h of demand. Optional F09–F10 are excluded from both figures.

Feasibility and scheduling notes

Availability baseline. PA0 set 8–10 h per member per week, rising to about 12 h in W8–W10. On 21/10/2026 all four members agreed to 12 h in W4–W5, 10 h in W6–W7, 13 h in W8, 20 h in W9–W10 and 8 h in W11. Estimates were not reduced to fit the hours; unused earlier time is not treated as available for later work.

Window	Agreed capacity	Allocated estimated effort	Capacity minus demand
W1–W3	10 h × 4 × 3 = 120	52 initiation/planning/tools + 5 monitoring = 57	+63
W4–W5	12 h × 4 × 2 = 96	75 requirements + 8 exploratory spike + 5 monitoring = 88	+8
W6–W7	10 h × 4 × 2 = 80	60 design (excluding W4 spike) + 5 monitoring = 65	+15
W8–W10	13 h × 4 + 20 h × 4 × 2 = 212	36 PA4 + 161 required PA5 + 7 monitoring = 204	+8

W11	8 h × 4 × 1 = 32	13 presentation and demo = 13	+19

4.4 Schedule

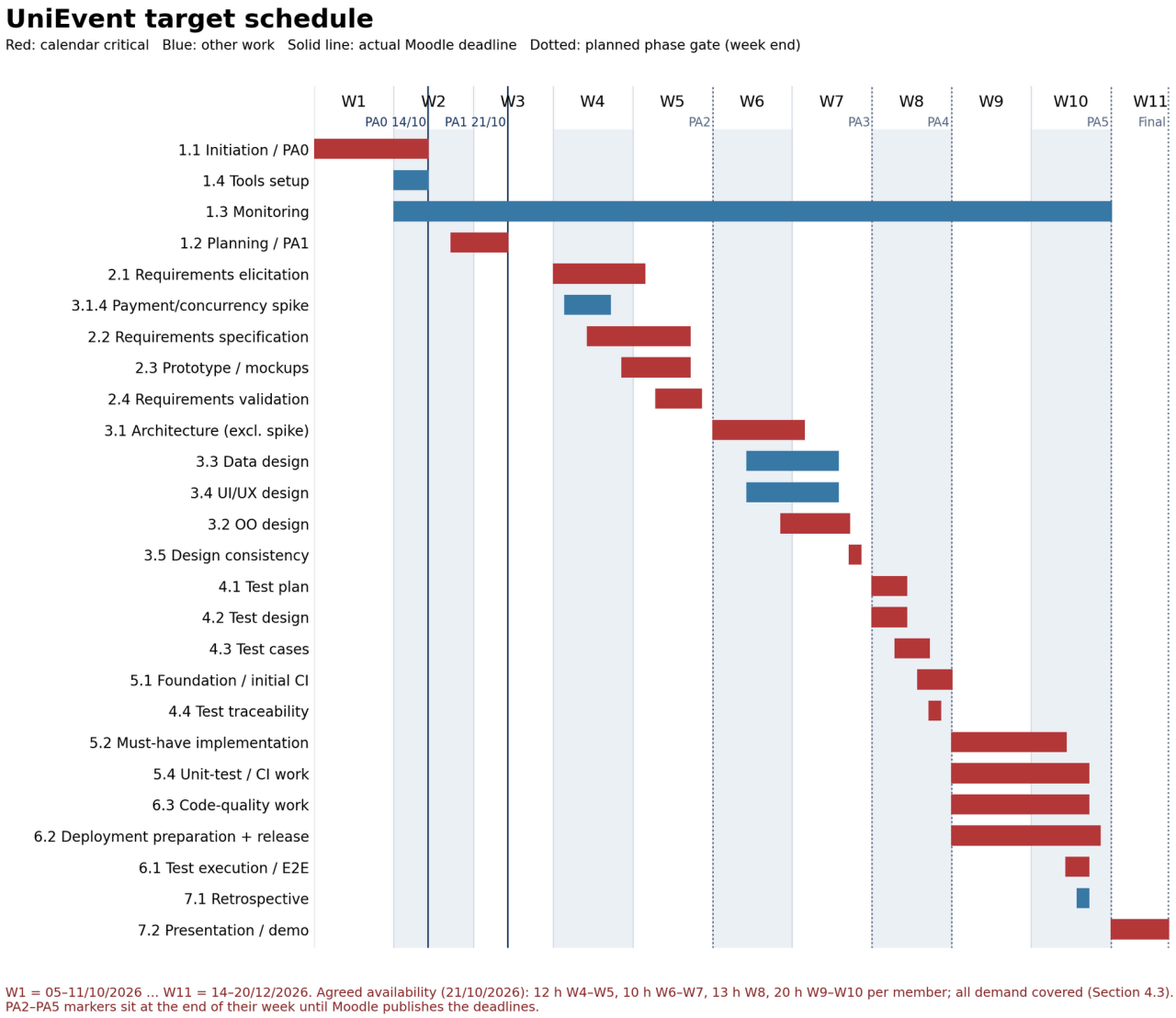


Figure 3. Target Gantt chart with actual PA0/PA1 deadlines and planned phase gates

Scheduling convention. W denotes the course week; weekdays form a relative Monday–Sunday planning calendar. FS/SS/FF mean finish-to-start, start-to-start and finish-to-finish; lags are calendar days. This relative schedule must be mapped to the actual Moodle deadline for every PA before baselining. Individual drafts finish at T−2 days, integration/review corrections and freeze

occur at T−1, and submission occurs at T. Saturday validation/release rows represent integration checks.

Table 4. Target schedule and dependencies (*calendar CPM only)

ID	Activity	Start	Finish	Predecessors	CPM critical*
1.1	Initiation / PA0	W1 Mon 05/10	W2 Wed 14/10	—	Yes
1.4	Tools setup	W2 Mon 12/10	W2 Wed 14/10	1.1 SS+7d	No
1.3	Monitoring	W2 Mon 12/10	W10 Sun 13/12	1.1 SS+7d	No
1.2	Planning / PA1	W2 Sat 17/10	W3 Wed 21/10	1.1 FS+2d; 1.4 FS	Yes
2.1	Requirements elicitation	W4 Mon 26/10	W5 Mon 02/11	1.2 FS+4d	Yes
3.1.4	Payment/concurrency spike	W4 Tue 27/10	W4 Fri 30/10	2.1 SS+1d; exploratory only	No
2.2	Requirements specification	W4 Thu 29/10	W5 Fri 06/11	2.1 SS+3d	Yes
2.3	Prototype / mockups	W4 Sun 01/11	W5 Fri 06/11	2.1 SS+6d	Yes
2.4	Requirements validation	W5 Wed 04/11	W5 Sat 07/11	2.2 SS+6d; 2.3 SS+3d; both FF+1d; 3.1.4 FS	Yes
3.1	Architecture (excl. spike)	W6 Mon 09/11	W7 Mon 16/11	2.4 FS+1d	Yes
3.3	Data design	W6 Thu 12/11	W7 Thu 19/11	3.1 SS+3d	No
3.4	UI/UX design	W6 Thu 12/11	W7 Thu 19/11	3.1 SS+3d	No

ID	Activity	Start	Finish	Predecessors	CPM critical*
3.2	OO design	W6 Sun 15/11	W7 Fri 20/11	3.1 SS+6d; 3.1 FF+4d	Yes
3.5	Design consistency	W7 Sat 21/11	W7 Sat 21/11	3.2 FS; 3.3 FS; 3.4 FS	Yes
4.1	Test plan	W8 Mon 23/11	W8 Wed 25/11	3.5 FS+1d	Yes
4.2	Test design	W8 Mon 23/11	W8 Wed 25/11	3.5 FS+1d	Yes
4.3	Test cases	W8 Wed 25/11	W8 Fri 27/11	4.1 SS+2d; 4.2 SS+2d; both FF+2d	Yes
5.1	Foundation / initial CI	W8 Fri 27/11	W8 Sun 29/11	3.5 FS+5d; foundation only, no feature delivery	Yes
4.4	Test traceability	W8 Sat 28/11	W8 Sat 28/11	4.3 FS	Yes
5.2	Must-have implementation	W9 Mon 30/11	W10 Wed 09/12	4.4 FS+1d; 5.1 FS	Yes
5.4	Unit-test / CI work	W9 Mon 30/11	W10 Fri 11/12	5.1 FS; 5.2 FF+2d	Yes
6.3	Code-quality work	W9 Mon 30/11	W10 Fri 11/12	5.1 FS; 5.2 FF+2d	Yes
6.2	Deployment preparation + release	W9 Mon 30/11	W10 Sat 12/12	5.1 FS; 6.1 FF+1d; 5.4 FF+1d; 6.3 FF+1d	Yes
6.1	Test execution / E2E	W10 Thu 10/12	W10 Fri 11/12	5.2 FS; 5.4 FF; 6.3 FF	Yes
7.1	Retrospective	W10 Fri 11/12	W10 Fri 11/12	6.1 SS+1d	No
7.2	Presentation / demo	W11 Mon 14/12	W11 Fri 18/12	6.2 FS+1d	Yes

Calendar critical-path analysis. A forward/backward pass over the durations and FS/SS/FF constraints in Table 4 gives zero calendar float for 1.1, 1.2, 2.1, 2.2, 2.3, 2.4, 3.1, 3.2, 3.5, 4.1, 4.2, 5.1, 4.3, 4.4, 5.2, 5.4, 6.3, 6.1, 6.2 and 7.2. Parallel critical branches exist for specification/mockups and test plan/test design. Tools setup (1.4) finished on 14/10 with two days of float; data design, UI/UX design, the W4 spike and the retrospective also have positive float. Zero float refers to the activity network with review and freeze time reserved, not to spare hours. The agreed availability in Section 4.3 covers the demand of this network.

Effort is not duration. For example, WBS 2.2 Requirements Specification requires 47 person-hours, while WBS 5.2 Must-have Features requires 94 person-hours. Their scheduled durations depend on parallel work, task dependencies, and confirmed team capacity rather than effort alone.

Schedule status. On 21/10/2026 all four members agreed to raise availability to 12 h/member/week in W4–W5, 13 h in W8 and 20 h in W9–W10, and to keep 10 h in W6–W7. With these hours no member-week in Section 4.5 is overloaded: W4–W5 provide 96 h for 88 h of demand and W8–W10 provide 212 h for 204 h. The added effort is a planning commitment, not an assumption of unlimited availability; R4 tracks it weekly.

4.5 Roles and Responsibilities

Role	Main	Backup	Responsibilities
Team lead / Project manager	A	C	Plan, assign tasks, weekly status report, contact point with the TA, Moodle submissions, presentation lead
Business analyst	B	A	Elicit and document requirements, organise reviews, traceability matrices, change log, AI usage log
Designer / Architect	C	D	Architecture, ADRs, class, data and UI design; consistency of diagrams and IDs across PAs
Developers	A, B, C, D	Module reviewer	Implement their modules (WBS 5.x), unit test, review other members' pull requests
Tester	D	B	Test plan, test cases, system testing, E2E and load tests, test report and defect log

Role	Main	Backup	Responsibilities
DevOps (CI/CD, deployment)	D	B	Repository settings, CI pipeline, deployment, monitoring during the demo

Table 5. Member codes used in this document

Code	Full name	Student ID
A	Lê Nguyễn Quốc Học	24127044
B	Thái Quang Huy	24127177
C	Trần Vũ Anh Kiệt	24127196
D	Nguyễn Văn Minh	24127205

Each row lists the work packages a member executes in that week. Every work package is executed only by its owner in Table 2, so no package appears under two members. "Monitoring" is the member's share of their own WBS 1.3 packages (A: 1.3.1–1.3.2; B: 1.3.3). Capacity is the availability agreed on 21/10/2026 (Section 4.3); no member-week is overloaded.

Weekly execution demand per member

Wk	Member	Work packages (WBS ID: hours)	Need h	Avail. h	Gap h
W4	A – Học	2.2.1: 4 h; 2.2.2: 4 h; monitoring: 2 h	10	12	+2
W4	B – Huy	2.1.1: 6 h; 2.1.2: 4 h; 2.2.5: 1 h; monitoring: 0.5 h	11.5	12	+0.5
W4	C – Kiệt	2.2.3: 5 h; 2.3.1: 4 h; 2.2.8: 2 h	11	12	+1
W4	D – Minh	3.1.4: 8 h; 2.2.4: 3 h	11	12	+1
W5	A – Học	2.2.1: 2 h; 2.2.2: 3 h; 2.2.7: 5 h; monitoring: 1.5 h	11.5	12	+0.5
W5	B – Huy	2.1.2: 1 h; 2.2.5: 4 h; 2.4.2: 5 h; monitoring: 1 h	11	12	+1

Wk	Member	Work packages (WBS ID: hours)	Need h	Avail. h	Gap h
W5	C – Kiệt	2.2.3: 1 h; 2.2.8: 2 h; 2.3.1: 4 h; 2.4.1: 4 h	11	12	+1
W5	D – Minh	2.2.4: 4 h; 2.2.6: 7 h	11	12	+1
W6	A – Học	3.2.2: 3 h; monitoring: 2 h	5	10	+5
W6	B – Huy	3.2.1: 4 h; 3.2.3: 2 h; monitoring: 0.5 h	6.5	10	+3.5
W6	C – Kiệt	3.1.1: 5 h; 3.1.2: 4 h; 3.4.1: 1 h	10	10	0
W6	D – Minh	3.1.3: 3 h; 3.3.1: 6 h	9	10	+1
W7	A – Học	3.2.2: 3 h; 3.5.1: 5 h; monitoring: 1.5 h	9.5	10	+0.5
W7	B – Huy	3.2.1: 3 h; 3.2.3: 3 h; monitoring: 1 h	7	10	+3
W7	C – Kiệt	3.1.2: 2 h; 3.4.1: 6 h	8	10	+2
W7	D – Minh	3.1.3: 3 h; 3.3.1: 2 h; 3.3.2: 5 h	10	10	0
W8	A – Học	4.2.1: 4 h; 4.3.1: 5 h; monitoring: 2 h	11	13	+2
W8	B – Huy	4.2.2: 4 h; 4.4.1: 4 h; 5.1.2: 4 h; monitoring: 1 h	13	13	0
W8	C – Kiệt	4.3.3: 6 h; 5.1.1: 6 h	12	13	+1
W8	D – Minh	4.1.1: 6 h; 4.3.2: 7 h	13	13	0

Wk	Member	Work packages (WBS ID: hours)	Need h	Avail. h	Gap h
W9	A – Học	5.2.2: 6 h; 5.2.3: 7 h; 5.2.8: 4 h; monitoring: 1.5 h	18.5	20	+1.5
W9	B – Huy	5.2.1: 8 h; 5.2.13: 5 h; 5.2.11: 5 h; monitoring: 0.5 h	18.5	20	+1.5
W9	C – Kiệt	5.2.4: 6 h; 5.2.5: 9 h; 5.2.9: 4 h	19	20	+1
W9	D – Minh	5.2.6: 8 h; 5.2.7: 9 h; 6.2.1: 3 h	20	20	0
W10	A – Học	5.2.8: 4 h; 6.2.2: 4 h; 6.3.1: 5 h; 7.1.1: 5 h; monitoring: 1.5 h	19.5	20	+0.5
W10	B – Huy	5.2.11: 3 h; 5.2.12: 4 h; 6.1.2: 7 h; 6.1.3: 5 h; monitoring: 0.5 h	19.5	20	+0.5
W10	C – Kiệt	5.2.9: 3 h; 5.2.10: 9 h; 5.4.1: 8 h	20	20	0
W10	D - Minh	5.4.2: 8 h; 6.1.1: 8 h; 6.2.1: 4 h	20	20	0

Ownership changes only through the task board and is recorded in the weekly report and the Revision History. Main and backup roles were agreed in the PA0 team contract and are unchanged.

4.6 Human Resources and Cost Estimate

The hypothetical labour rate is VND 70,000 per person-hour; labour contingency is 10%. Tools and deployment are budgeted on eligible no-cost plans, subject to quota and compatibility checks. The required scope includes a VND 100,000 reserve for authorised live payment testing. This is a planning allowance, not a quoted gateway fee or money already spent. The mock adapter used in CI has no transfer expense. Học records approval before any paid service or real-payment test.

Table 6. Required-scope estimate (F01–F08; F09/F10 excluded).

Item	Basis	Cost (VND)
Labour	427 h × 70,000	29,890,000
Tools	GitHub, Discord, Figma/draw.io: free plans	0
Hosting / static analysis	Eligible no-cost plans; quotas checked at setup	0
Payment test / demo budget	Reserve for live tests on the authorised channel	100,000
Contingency	10% of labour	2,989,000
REQUIRED-SCOPE TOTAL		32,979,000

Table 7. Alternative estimate if optional F09/F10 are also implemented.

Item	Basis / assumption	Cost (VND)
Labour (including optional)	443 h × 70,000	31,010,000
Tools / hosting / static analysis	As above	0
Payment test / demo budget	Reserve	100,000
Contingency	10% of labour	3,101,000
FULL-SCOPE TOTAL		34,211,000

The VND 100,000 test allowance is included once in each alternative budget. It is a provisional reserve, not a confirmed gateway fee. Returned test transfers are not net project expenditure; record unrecovered fees or test purchases. If eligible hosting, required live-channel access or the reserve is insufficient, Học seeks an explicit revised budget/delivery decision before spending.

Cost estimates measure hypothetical economic effort. Capacity is handled separately in the schedule-status decision in Section 4.3.

5 Risk Management

Risk assessment. Probability P: 0.1 (very unlikely) to 0.9 (very likely); loss L: 1 (negligible) to 10 (project failure). RE = P × L. Risks are ordered by descending RE; IDs remain stable for tracking. Review RE ≥ 3.0 weekly and other risks at each phase gate, or sooner if triggered. Owners record actions, due dates and revised scores in weekly reports.

Risk ownership follows the PA0 roles: Học (PM), Huy (BA), Kiệt (Architect) and Minh (Tester/DevOps); all four also develop software. Risk scores are planning assessments, reviewed weekly or when triggered. Minh executes the W4 payment/concurrency spike (WBS 3.1.4), and Kiệt reviews its design and results.

ID	Risk	Type	Probability	Loss	RE	Mitigation / contingency	Owner
R4	Failure to sustain the increased weekly availability agreed in Section 4.3, or member absence, may delay delivery or compress Must-have testing and documentation.	People / estimation	0.6	8	4.8	Mitigation: each member logs actual hours on GitHub issues; weekly reports compare them with the agreed 12 h in W4–W5, 13 h in W8 and 20 h in W9–W10. PA5 foundation work (WBS 5.1) starts after the PA3 gate. Contingency: if a member misses the agreed hours for two consecutive weeks, Học escalates to the TA and the team records a scope or schedule decision. Preserve Must-have tests and continue weekly risk monitoring.	Lê Nguyễn Quốc Học

ID	Risk	Type	Probability	Loss	RE	Mitigation / contingency	Owner
R1	A member cannot explain AI-generated code or text, allowing defects or unsupported claims into a submission.	People / quality	0.5	8	4.0	Mitigation: require owner and peer verification, AI logs and walkthroughs. Contingency: block unexplained work from submission, pair-review and rewrite it; repeat relevant checks and update the log with actual verification.	Thái Quang Huy
R2	Duplicate callbacks or concurrent purchase/expiry processing issue duplicate tickets or exceed event capacity.	Technology	0.4	9	3.6	Mitigation: W4 tests cover concurrency, repeated callbacks and late payments; use atomic capacity updates, idempotency and unique ticket issuance. Contingency: pause affected sales; flag exceptions as pending reconciliation, retaining evidence and preventing duplicate processing. For paid tickets, issue only after verifying payment and available capacity. Free tickets bypass payment but remain subject to capacity and duplicate-issuance controls. Kiệt coordinates technical resolution; Học liaises with the organiser/account holder. Record the completed resolution and payer notification before closure; stopping issuance alone is not resolution.	Trần Vũ Anh Kiệt

ID	Risk	Type	Probability	Loss	RE	Mitigation / contingency	Owner
R3	Delayed payOS access or invalid/missing webhooks prevent payment confirmation and QR ticket delivery.	Technology	0.4	8	3.2	Mitigation: Minh verifies one authorised payOS channel, signatures, callback matching and expiry in the W4 spike (WBS 3.1.4), and Kiệt reviews the spike result. Học confirms the account holder, TA permission, exception ownership and payer communication before transfers. Contingency: use a clearly labelled mock for continued development/CI, retain payment evidence and escalate blocked live acceptance. Mock results do not complete the live-channel commitment; a changed PA5 boundary requires an explicit TA/team decision. No automated refunds or custody of funds are included.	Trần Vũ Anh Kiệt
R5	Exposed secrets or participant records enable unauthorised access or disclosure of personal data.	Security / privacy	0.3	9	2.7	Mitigation: exclude secrets from Git and AI prompts; use synthetic data and review access controls. Contingency: revoke credentials, contain access, assess affected records and notify the team lead; remove exposed data and review the cause.	Nguyễn Văn Minh

ID	Risk	Type	Probability	Loss	RE	Mitigation / contingency	Owner
R6	Hosting, email or another external service outage interrupts ticket delivery or the live demo.	Technology	0.3	8	2.4	Mitigation: check service health and rehearse a local demo with synthetic data; prepare a backup video. Contingency: identify the fallback as such, restore service and retry email delivery for the same ticket, without creating another entitlement.	Nguyễn Văn Minh
R7	Lost connectivity, denied camera access or incompatible devices stop QR check-in and cause queues.	Technology	0.3	7	2.1	Mitigation: test HTTPS camera access and supported phones; prepare a backup device and network. Contingency: switch device/network; pause admissions until server validation returns, then reject invalid or used tickets. Do not bypass validation with offline admission.	Nguyễn Văn Minh
R8	New requests or Should-have work consume time needed to implement and test F01–F08.	Scope	0.4	5	2.0	Mitigation: assess each change against PA0, effort and dependencies with the team lead. Contingency: defer F09–F10 or new requests, record the decision and revise the plan while preserving Must-have implementation, tests and review.	Thái Quang Huy
R9	Requirements based on limited survey sources miss other clubs' workflows, causing unsuitable rules and rework.	Requirements	0.4	4	1.6	Mitigation: compare other clubs' event information and interview additional organisers in PA2; record assumptions. Contingency: revise affected requirements and traceability, obtain stakeholder feedback and agree a bounded workflow before implementation.	Thái Quang Huy

Shaded rows have RE ≥ 3.0 and require weekly review; other risks are reviewed at phase gates or sooner if triggered. The revised availability covers planned demand; R4 remains open to monitor delivery against that commitment. F01–F08 remain Must-have; F09–F10 remain Should-have.

6 Tools & Collaboration Setup

PA0 working rules remain in force. The verification record below distinguishes configuration observed on 21 October 2026 from checks that require team or TA confirmation. Repository evidence was read directly; README statements alone are not treated as proof of enforcement. Application CI is a later WBS 5.1.2 milestone.

- ✔ **Moodle** – course announcements and PA submissions. Lê Nguyễn Quốc Học submits the required files; Trần Vũ Anh Kiệt is the PA0 backup. Retain the submitted version, source and submission confirmation according to course requirements.
- ✔ **Communication – Discord** server "UniEvent – T07": https://discord.gg/zVMjhEGnzp. The TA confirmed that membership in the team's Discord server is not required. Use Discord for discussion and reviews; record decisions and action owners in weekly reports and linked issues. Respond within 12 hours on weekdays and 24 hours at weekends.
- ✔ Task board – GitHub Projects "Team Management": https://github.com/orgs/UniEvent-T07/projects/1. PA1 issues #29–#32 cover Sections 2–6. Học confirms one assignee and the actual due date for each issue before submission; due dates for #31 and #32 remain pending verification. From W4, each Level-3 work package is to have an issue matching Table 2: WBS ID, one executing owner, estimate, week, output, due date, status and evidence links. Học checks full WBS coverage before submission and maintains it weekly. States: Backlog, To do, In progress, In review, Done. Shared-document review evidence is acceptable for PA1 section tasks.
- ✔ Git repository – https://github.com/UniEvent-T07/UniEvent. The repository and default branch main were verified on 21 October 2026. README, .gitignore, issue/PR templates and the core documentation/source folders exist. The PA1 application stack is selected as

in Section 3.2. Học records the instructor/TA account and confirms repository access before submission; this check remains pending.

- The active "Protect main" ruleset was verified on 21 October 2026: it targets the default branch, requires a pull request with one approving review, dismisses stale approvals after new commits, requires resolution of review threads, and blocks deletion and non-fast-forward updates. It has no listed bypass actors. Evidence: https://github.com/UniEvent-T07/UniEvent/rules/25001815. Application lint/build/test checks are not in the observed ruleset; Huy adds the CI and required status checks under WBS 5.1.2 before the first Must-have implementation merge, then records a failing PR blocked and a passing PR accepted.
- Proposed branch convention: type/issue-short-description; PRs link the task, relevant requirement IDs and verification evidence. Definition of Done for documents: complete content, review by another member, resolved feedback and updated traceability. For application code: acceptance criteria met, appropriate tests recorded, peer approval, required CI checks passed and related documentation updated. Manual checks during setup do not waive the CI milestone for Must-have implementation; if it is missed, keep the PR unmerged and escalate to Học.
- README, .gitignore and issue/PR templates were observed in main on 21 October 2026. The .gitignore excludes .env/.env.*, keys, build output and test artefacts while permitting .env.example. Add sample-only .env.example files with the application foundation under WBS 5.1.2. Real credentials remain outside Git and AI prompts; use deployment secret settings and synthetic test data. The observed issue-to-Discord workflow is notification automation and does not establish application CI or current Discord delivery.
- Use Conventional Commits (feat:, fix:, docs:, test:, refactor:, chore:). PA1 owners edit the shared document and reviewers leave comments/suggestions; separate section PRs are unnecessary. Finish individual work 2 days before the PA deadline; the lead

integrates and freezes the reviewed version 1 day before it. Commit the final PDF and source to /pa/PA1 through a reviewed PR before Moodle submission.

- Record significant AI use in /docs/ai-usage/PAx.md, including this PA1 revision; Thái Quang Huy consolidates entries. Each owner records the actual purpose, human checks and corrections, and must explain the result; another member verifies it. Keep the phase log consistent with the document AI Usage Log. Do not claim review merely because AI suggests it; exclude secrets, real payment data and third-party personal data.

✔ **Weekly status meeting** with the TA: Thursday 20:00–20:15 (attendance to confirm). Internal meetings: Tuesday 21:00–21:45 and Saturday 21:00–21:30. Submit the **weekly status report** for W2–W10 to /docs/management/weekly/W<week>-status.md. **Deadline: Sunday 22:00 (GMT+7).** All meeting times are GMT+7. Record progress, blockers, risks and decisions; preserve previous reports and document corrections in the next report. Give at least 12 hours notice of absence. Seek consensus, then vote after 24 hours if needed; the lead breaks ties.

Evidence owners and deadlines: Học closes the PA1 contribution confirmation, board coverage/due-date checks and TA access/membership checks before submission. Huy maintains the repository/ruleset record and provides application CI evidence at WBS 5.1.2; Minh owns deployment evidence at WBS 6.2.1. Huy reconciles the AI log. An open item is not marked complete without direct evidence; invitations alone do not prove membership. Keep the setup record and linked PA1 issues consistent.

PA1 setup verification record — checked 21 October 2026

Evidence links: main repository tree | main protection | W3 status report

Item	Status	Evidence / next action	Check / deadline
Repository and initial files	Verified	main tree, README, .gitignore and issue/PR templates	21 Oct; connector read
Main protection / peer review	Verified	Active Protect main ruleset, #25001815; one approval; no bypass actors	21 Oct; connector read

Item	Status	Evidence / next action	Check / deadline
PA1 tasks / full WBS coverage	Partial	Học confirms PA1 assignees/due dates and full WBS coverage; Level-3 issue details follow Table 2.	Before submission
TA Discord membership	– Not required	Invite URL is recorded; current delivery and TA membership need direct confirmation (Học)	Before submission
Instructor/TA repository access	Not required	Học records the actual accounts and access check; no account is assumed	Before submission
Weekly reports	Files verified	W2-status.md and W3-Status.md exist; timeliness not independently checked	21 Oct; connector read
Application quality CI	Planned	No lint/build/test workflow in the inspected main tree; Huy implements WBS 5.1.2	Before first feature merge

Repository structure: existence checked in main on 21 October 2026; delivery contents follow their phase gates

Folder	Content
/src	Exists: /src/README.md. Application code is added with WBS 5.1; planned layout /src/web and /src/api follows the selected stack.
/docs/management	Exists: /docs/management/weekly, including W2-status.md and W3-Status.md. Học standardises W3-Status.md to W3-status.md and updates links after the rename.
/docs/devops	Exists: task-notifications.md describing the issue-to-Discord automation.
/docs/guides	Exists: Vietnamese team guides for Discord and GitHub Projects.
/docs/requirements	Exists with README. PA2 adds the SRS, use cases, mockups, glossary, traceability and interview notes; Huy checks the PA2 gate.
/docs/design	Exists with README. PA3 adds C4/UML models, ERD and ADRs; Kiệt checks the design baseline.
/docs/test	Exists with README. PA4 adds test plans/cases; PA5 adds execution reports and defect evidence; Minh checks each gate.
/docs/ai-usage	Exists: PA0.md and PA1.md. Huy reconciles the repository PA1 log with this document before submission.

Folder	Content
/pa/PA1 ... /pa/PA5	/pa/PA0 exists with PDF/source. /pa/PA1 through /pa/PA5 were not present in the inspected tree. Học creates each phase folder and commits its matching reviewed PDF/source before submission.
/.github	Exists: issue template, PR template, team.json, scripts/issue-to-discord.mjs and the issue-discord.yml notification workflow. Huy implements required application lint/build/test CI under WBS 5.1.2.
/scripts	Root-level helper scripts are optional; the existing notification script is in /.github/scripts. The implementing owner documents and peer-reviews scripts before use.

7 AI Usage Log

#	Member	Tool	Task / purpose	How the output was verified and changed
1	Lê Nguyễn Quốc Học	ChatGPT (OpenAI)	Assisted in planning and drafting Section 4 (Project Plan), including WBS, effort estimation, schedule, roles, and cost analysis.	Reviewed AI-generated planning suggestions against the course rubric and PA0. Cross-checked WBS IDs, effort estimates, dependencies, and team capacity; identified inconsistencies for revision.
2	Lê Nguyễn Quốc Học	ChatGPT (OpenAI)	Reviewed PA1 against the course rubric, PA0 baseline, and official template to identify missing content and inconsistencies.	AI identified discrepancies in WBS IDs, effort estimates, and schedule dependencies. Findings were marked for manual verification and correction.
3	Lê Nguyễn Quốc Học	ChatGPT (OpenAI)	Evaluated the consistency and feasibility of the WBS, effort estimates, project schedule, and resource allocation.	AI highlighted inconsistent work-package references, dependency issues, and a capacity shortfall. Recommendations were considered for revising the project plan.

#	Member	Tool	Task / purpose	How the output was verified and changed
4	Lê Nguyễn Quốc Học	ChatGPT / Codex (OpenAI)	21 October 2026: revised PA1 text, payment boundary, capacity reconciliation, contributor hours, target schedule/Gantt, setup evidence and costs; prepared the updated Word file for the original Drive record.	User confirmed the selected stack, one live payOS channel and retention of PA0 hours. AI checked WBS/resource sums, read repository/ruleset/report evidence and rendered the revised layout. All-member percentages, TA access, merchant authorisation and capacity recovery remain pending human confirmation; no additional completed review is claimed.
5	Nguyễn Văn Minh	ChatGPT / Codex (OpenAI)	Drafted and iteratively revised Sections 5–6 using PA0 and the PA1 template; compared V3 with earlier drafts; reviewed risk scoring/owners, payment exceptions and collaboration/CI rules. On 21 October 2026, checked alignment with the updated Section 4 and prepared this log entry.	Minh supplied references, clarified ownership and formatting constraints, and requested successive reviews and corrections. Revisions distinguished free/paid tickets, clarified payment-exception closure and CI deadlines, and preserved the template. AI recalculated RE, checked WBS/capacity links and rendered/compared the DOCX. The latest check matched the 8 h W4–W5 and 55 h W8–W10 deficits to R4. Human/peer approval and capacity recovery remain pending; this entry does not claim that Minh tested the application or configured repository/Discord settings through these sessions.

#	Member	Tool	Task / purpose	How the output was verified and changed
6	Thái Quang Huy	ChatGPT (OpenAI)	Used ChatGPT to review the UniEvent PA1 proposal against the original document, GitHub repository, PA1 template, and course guidelines. Requested checks of the Vision, proposed architecture, feature scope, WBS, effort estimates, Gantt schedule, resource feasibility, cost calculations, risks, and collaboration setup. ChatGPT identified inconsistencies, suggested corrections, and produced separate revised reference documents.	Requested multiple review iterations instead of accepting the first output. ChatGPT cross-checked WBS identifiers, effort totals (422 required hours and 438 hours including optional work), cost calculations, and repository evidence. Unresolved issues, including schedule feasibility and contribution evidence, were explicitly identified for team confirmation.
7	Thái Quang Huy	ChatGPT (OpenAI)	Used ChatGPT to analyse 22 historical PA1 evaluation comments from Intro2SE class 24C05 and compare them with UniEvent PA1 and the course grading rubric. Requested improvements to actor-specific problem analysis, stakeholder responsibilities, access permissions, data ownership, work-package accountability, revision tracking, and completeness of required sections. ChatGPT prepared the PA1 V2 reference document and a feedback-comparison report.	Requested further checks against the original PA1 template and grading criteria. ChatGPT revised the proposed content to address relevant historical feedback and identified remaining limitations. Stakeholder assumptions, individual contributions, and the project's critical path were not treated as independently validated.

#	Member	Tool	Task / purpose	How the output was verified and changed
8	Thái Quang Huy	ChatGPT (OpenAI)	Used ChatGPT to perform an additional review of PA1 V2, refine the content, and prepare PA1 V3 as a reference for team members to copy into the shared PA1 document. Clarified that the team collaboratively edits one Google Drive document and tracks PA1 through GitHub Project issues rather than separate pull requests for each section. Requested formatting alignment with the original template, updated tables, and a section-by-section handoff guide.	ChatGPT compared the revised document with the original template, checked the exported document structure (26 tables and 3 embedded figures), reviewed pagination, and updated the reference document's table of contents. It also highlighted the unresolved W9–W10 resource deficit, Gantt/critical-path validation, and pending contribution and AI-log evidence. The original shared PA1 document was not overwritten by ChatGPT; final content acceptance and submission remain the team's responsibility.
9	Thái Quang Huy	Claude (Anthropic)	21/10/2026: reviewed PA1 against 22 historical PA1 comments from class 24C05, PA0 and the repository; proposed one-owner work allocation, per-problem and per-user Vision tables, organisation administration (F01, WBS 5.2.13), calendar dates and corrected costs; regenerated Figure 3.	Claude recomputed every WBS, capacity and cost total with scripts and checked each work package against the weekly allocation. The member checked each change against the template; the capacity option (B) and contribution percentages were decided by the whole team on 21/10/2026.